CH 4 - Multivariate Discrete Random Variable

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4.1 Basic Terminology and Techniques

Ofter we are interested in multiple phenomenal that vary alongside each other.

Examples:

- Treatment/dose and disease status
- Stock price or credit rating of 2 companies
- Pollutant level and respiratory illnesses

> 📘 **Definition — Joint Probability Mass Function**
>
> The **joint probability mass function** of two random variable X and Y is
>
> $$f(x,y) = P(X = x, Y = y)$$
>
> for (x, y) in joint rnage of X, Y respectively.
>
> It is represented usually by a table, occasionally a function, never a histogram.

Properties for JPMF:

1. $0 \leq f(x,y) \leq 1$
2. $\sum_x \sum_y f(x,y) = 1$

We can find any probability involving x and y by summing up the $f(x,y)$ over all pairs that satisfy

Example:

Flip a fair coin 3 times and let $X = \#$ of heads obtained, $Y = \begin{cases} 1 \text{ if first flip is T} \\ 0 \text{ else} \end{cases}$. Find $f(x,y)$

	0	1	2	3
0	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$
1	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0

$P(2 < X + Y) = f(3,0) + f(2,1) + f(3,1) = \frac{2}{8}$

$P(X - Y = 1) = f(1,0) + f(2,1) = \frac{2}{8}$

> 📘 **Definition — Marginal PMF**
>
> The marginal PMF of X is
>
> $$f_X(x) = P(X = x) = \sum_y f(x,y)$$
>
> $$f_Y(y) = P(Y = y) = \sum_x f(x,y)$$

In general, the JPMF of $X_1, X_2, ..., X_n$ is $f(x_1, x_2, ..., x_n) = P(X_1 = x_1, X_2 = x_2, ..., X_n = x_n)$

Marginal PMF of X_1 is $f_1(x_1) = \sum_{X_2} ... \sum_{X_n} f(x_1, x_2, ..., x_n)$.

The JPMF of X_1, X_2 is $f_{1,2}(x_1, x_2) = \sum_{X_3} ... \sum_{X_n} f(x_1, x_2, ..., x_n)$

> 📘 **Definition — Independence Random Variable**
>
> We say two random variables X, Y are independent if and only if
>
> $$f(x, y) = P(X = x, Y = y) = P(X = x)P(Y = y) = f_X(x) + f_Y(y)$$

Example:

Let $X = \#$ days between texts from A to B and $Y = \#$ days between texts from B to A

The joint PMF be $f(x, y) = \frac{1}{16}(\frac{1}{2})^x(\frac{7}{8})^y$, for $x, y = 0, 1, 2, ...$

Find the marginal PMF:

$f_X(x) = \frac{1}{16}(\frac{1}{2})^x \sum_y (\frac{7}{8})^y = \frac{1}{16}(\frac{1}{2})^x 8 = (\frac{1}{2})^{x+1}$

$X \sim \text{Geo}(\frac{1}{2})$

$f_Y(y) = \frac{1}{16}(\frac{7}{8})^y \sum_x (\frac{1}{2})^x = \frac{1}{16}(\frac{7}{8})^x 2 = \frac{1}{8}(\frac{7}{8})^y$

$Y \sim \text{Geo}(\frac{1}{8})$

$f(x, y) = f_X(x)f_Y(y) = (\frac{1}{2})^{x+1}\frac{1}{8}(\frac{7}{8})^y = \frac{1}{16}(\frac{1}{2})^x(\frac{7}{8})^y$, thus X, Y are independent

4.2 Functions of 2^+ random variables

We may be interested in $U = g(X, Y)$

Examples:

1. Let $X = \#$ marks gained, $Y = \#$ marks lost

$y \setminus x$	0	1	2	3
0	0.18	0.34	0.03	0.02
1	0.09	0.09	0.06	0.08
2	0.01	0.01	0.06	0.03

   If U is the overall impact, $U = g(X, Y) = X - Y$

   PMF of U:

u	-2	-1	0	1	2	3
	0.01	0.1	0.33	0.43	0.11	0.02

   In general, if $U = g(X, Y)$, $f_U(u) = \sum_x \sum_y f(x, y)$

   In the case of $U = g(X, Y) = X + Y$

   $f_U(u) = \sum_x \sum_y f(x, y) = \sum_x^u f(x, u - x)$

2. Suppose $X \sim \text{Poi}(7), Y \sim \text{Poi}(5)$ are independent

   $U = X + Y$

   $f_U(u) = f_X(x)f_Y(y) = \frac{e^{-7}7^x}{x!} \cdot \frac{e^{-5}5^y}{y!} = \sum_{x=0}^{u} f(u, u-x) = \frac{1}{e^{12}} \sum_{x=0}^{u} = \frac{7^x 5^{u-x}}{x!(u-x)!}$

   $= \frac{e^{-12}}{u!} \sum_{x=0}^{u} = \binom{u}{x} 7^x 5^{u-x} = \frac{e^{-12}12^x}{u!}$

   $U \sim \text{Poi}(12)$

Poisson random variables are closed under convolution:

$X \sim \text{Poi}(x), Y \sim \text{Poi}(y)$ are independent, $X + Y \sim \text{Poi}(x + y)$

Binomial random variables are closed under convolution:

$X \sim \text{Bin}(n, p), Y \sim \text{Bin}(m, p)$ are independent, $X + Y \sim \text{Bin}(n + m, p)$

Geometric random variables are closed under convolution:

$X \sim \text{Geo}(p), Y \sim \text{Geo}(p)$ are independent, $X + Y \sim \text{NB}(p)$

> 📘 **Definition — Multinomial Random Variable**
>
> Trials are conducted with 3 conditions:
>
> - k possible outcomes on each Trials
> - probabilities are constant, $p_1, p_2, ..., p_k$
> - trials are independent
>
> We conduct n trials and let $X_i = \#$ of outcome i observed, $1 \leq i \leq k, k \in \mathbb{N}_{>0}$
>
> $X_1, .., X_k \sim \text{MN}(n, p_1, ..., p_k)$ where $\sum_{i=0}^{k} p_i = 1$ and $\sum_{i=1}^{k} X_i = n$
>
> $X_1, ..., X_{k-1} \sim \text{MN}(n, p_1, ..., p_{k-1})$
>
> **WARN**: trials are independent, but variables are not independent

Examples of :

- # of votes for candidates in election
- # outcomes on a dice roll
- # of each answer to survey/question

Suppose roll 10 fair 6-sided dices and let

$X_1 = \#$ of $2, X_2 = \#$ of $6, X_3 = \#$ of 2

What is the probability of $f(2, 6, 2) = \left(\frac{10!}{2!6!2!}\right)\left(\frac{1}{6}\right)^2\left(\frac{1}{3}\right)^6\left(\frac{1}{2}\right)^2$

Joint PMF:

$$f(x_1, ..., x_k) = \frac{n!}{x_{1!}...x_{k!}} p_1^{x_1}...p_k^{x_k}, \ \forall \sum_{i=1}^{k} x_i = n$$

$$f(x_1, ..., x_{k-1}) = \frac{(n-1)!}{x_{1!}...x_{(k-1)!}\left(n - \sum_i^{k-1} x_i\right)!} p_1^{x_1}...p_{k-1}^{x_{k-1}} \left(\sum_i^{k-1} x_i\right)^{n - \sum_i^{k-1} x_i}, \ \forall \sum_{i=1}^{k} x_i = n$$

Marginal PMF:

$$f_{X_1}(x_1) = \sum_{x_2} \dots \sum_{x_k} f(x_1, \dots, x_k)$$

Example, $k = 3$

$f_{X_1}(x_1) = \sum_{x_2=0}^{n-x_1} f(x_1, x_2) = \sum_{x_2=0}^{n-x_1} \frac{n!}{x_1! x_2! (n-x_1-x_2)!} p_1^{x_1} p_2^{x_2} (1-p_1-p_2)^{n-x_1-x_2}$

$= \binom{n}{x_1} p_1 x^1 \sum_{x_2=0}^{n-x_1} \binom{n-x_1}{x_2} p_2^{x_2} (1-p_1-p_2)^{n-x_1-x_2} = \binom{n}{x_1} p_1^{x_1} (1-p_1)^{n-x_1}$

$X_1 \sim \text{Bin}(n, p_1), X_i \sim \text{Bin}(n, p_i)$

If $U = X_1 + X_2, k = 3$

$f_U(u) = \sum_{x_1=0}^{u} f(x, u-x_1) = \sum_{x_1=0}^{u} \frac{n!}{x_1!(u-x_1)!(n-u)!} p_1^{x_1} p_2^{u-x_1} (1-p_1-p_2)^{n-u}$

$= \binom{n}{u} (p_1+p_2)^u (1-p_1-p_2)^{n-u}$

4.3 Expectation, Covariance, Linear Correlation

Expectation

To find the expected value of a funciton of X and Y,

we let $U = g(X, Y)$ and find $E[U]$, $E[g(X,Y)] = E[U] = \sum_u u f_U(u) = \sum_x \sum_y g(x,y) f(x,y)$

Properties:

1. Linearity: $E[ag(X,Y) + bh(X,Y) + c] = aE[g(X,Y)] + bE[h(X,Y)] + c$
2. Independence: $E[g(X)h(Y)] = E[g(X)]E[h(y)]$
3. $(X_1, X_2, \dots, X_N) \sim \text{MN}(n, p_1, p_2, \dots, p_n)$

   For $n \geq 2$

   $E[X_i X_j] = n(n-1)p_1 p_2$

> 📘 **Definition — Covariance**
>
> The **covariance** of two random variables X, Y is
>
> $$\text{Cov}(X,Y) = \sigma_{XY} = E[(X-\mu_x)(Y-\mu_Y)] = E[XY] - \mu_X \mu_Y$$
>
> If $\text{Cov}(X,Y) > 0$, indicates positive **linear** association
>
> If $\text{Cov}(X,Y) < 0$, indicates negative **linear** association

> 📐 **Theorem — Independence and Covariance**
>
> If X, Y are independent random variables, then
>
> $$\text{Cov}(X,Y) = 0$$
>
> **WARN** the converse is FALSE
>
> Zero covariance DOES NOT imply independence.

Definition — Correlation

$$\text{Corr}(X, Y) = \rho_{XY} = \left| \frac{\text{Cov}(X, Y)}{\sigma(X)\sigma(Y)} \right| \leq 1$$

We can interpret the magnitude as the strengh of the linear relationship. The bigger the absolute value, the stronger the relation.